

KIRENSK, Russian Federation

WMO: 302300

Lat: 57.773N

Lon: 108.059E

Elev: 848

StdP: 14.25

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-50.1	-45.6	-54.3	0.2	-50.0	-49.7	0.3	-45.6	17.6	7.1	15.6	7.2	0.4	210	0.663

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.1	85.8	66.1	82.3	65.2	78.7	63.9	69.2	79.9	67.5	77.8	65.6	75.6	5.1	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.7	97.9	73.6	63.8	91.3	72.1	61.9	85.4	70.2	34.0	80.1	32.5	77.9	31.0	75.8	79.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	16.2	14.0	12.2	DB	-53.2	91.7	4.9	2.7	-56.8	93.6	-59.6	95.2	-62.4	96.8	-66.0	98.7
(5)				WB	-53.6	74.2	5.2	2.8	-57.4	76.3	-60.5	78.0	-63.4	79.6	-67.2	81.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	26.4	-15.5	-6.8	11.0	30.8	45.5	60.6	65.5	59.8	44.9	28.5	3.3	-13.1
	DBStd	31.13	18.11	16.00	13.74	8.82	7.44	7.27	5.80	6.34	7.26	9.84	18.58	19.15
	HDD50	9812	2029	1592	1208	577	181	7	0	7	187	666	1401	1957
	HDD65	14237	2494	2012	1673	1026	606	171	65	183	603	1131	1851	2422
	CDD50	1193	0	0	0	1	42	323	482	311	34	0	0	0
	CDD65	143	0	0	0	0	1	38	82	22	0	0	0	0
	CDH74	1956	0	0	0	0	46	650	937	314	9	0	0	0
	CDH80	550	0	0	0	0	7	187	281	75	0	0	0	0
Wind	WSAvg	4.4	3.2	3.5	4.6	5.7	5.9	4.8	4.1	3.8	4.5	5.1	4.5	3.3
Precipitation	PrecAvg	15.0	0.7	0.6	0.4	0.6	1.3	1.9	2.6	2.4	1.6	1.1	1.1	1.0
	PrecMax	20.2	1.5	1.3	1.0	1.7	3.2	4.0	7.1	4.6	2.8	2.7	1.8	1.9
	PrecMin	10.9	0.2	0.2	0.0	0.1	0.3	0.2	0.4	0.3	0.0	0.0	0.4	0.4
	PrecStd	2.4	0.3	0.2	0.2	0.3	0.6	0.9	1.4	1.1	0.6	0.5	0.4	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.7	32.2	47.4	61.1	81.2	89.1	90.9	86.5	74.9	54.7	40.8	28.7
		MCWB	24.8	28.4	36.0	44.9	57.5	65.9	69.1	67.1	59.8	45.3	37.5	26.7
	2%	DB	22.0	27.8	40.0	55.5	73.8	85.2	86.6	81.5	66.7	49.2	36.6	24.4
		MCWB	20.4	24.9	32.3	42.0	53.4	64.7	68.0	65.7	55.0	42.2	33.2	22.8
	5%	DB	16.9	23.2	35.2	50.1	68.1	81.6	82.8	77.1	61.1	44.9	33.5	20.4
		MCWB	15.6	21.0	29.6	39.1	50.4	63.6	66.3	64.1	52.5	39.8	31.4	19.1
	10%	DB	11.5	17.4	31.8	45.4	62.5	77.4	79.4	73.1	56.8	40.8	29.6	15.7
		MCWB	10.5	15.7	27.2	36.4	47.8	61.6	65.6	62.0	50.4	36.8	27.6	14.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.4	28.5	36.9	46.5	58.8	69.9	72.8	70.0	61.4	46.5	38.4	27.3
		MCDB	26.7	31.6	45.5	58.2	76.9	82.5	84.5	80.6	71.8	51.7	40.4	28.8
	2%	WB	20.5	24.9	33.1	42.8	55.3	67.1	70.2	67.4	57.1	43.5	33.8	22.9
		MCDB	22.0	27.6	38.9	54.2	69.9	79.9	81.1	77.7	65.0	48.1	35.9	24.3
	5%	WB	15.6	20.8	30.1	40.0	52.0	65.2	68.6	65.3	53.9	40.3	31.3	19.2
		MCDB	16.9	23.0	35.1	49.0	64.0	77.6	78.7	75.0	59.0	44.1	33.7	20.3
	10%	WB	10.5	15.8	27.2	37.3	49.3	63.2	67.0	63.1	51.1	37.3	27.6	14.5
		MCDB	11.6	17.5	31.6	44.2	60.9	74.6	76.4	71.3	56.0	40.6	29.6	15.6
Mean Daily Temperature Range	5% DB	MDBR	18.0	23.5	28.6	21.6	22.8	23.8	21.1	20.8	17.2	14.4	15.7	16.5
		MCDBR	17.6	23.0	28.5	28.7	34.8	32.2	30.0	29.5	26.4	20.5	13.0	16.2
		MCWBR	17.0	20.5	21.5	16.5	16.7	14.0	13.9	14.6	14.8	13.8	11.2	15.5
	5% WB	MCDBR	18.3	23.1	27.3	26.0	29.9	27.7	24.9	26.0	22.1	18.3	12.6	16.8
		MCWBR	17.7	20.8	21.0	15.4	15.3	13.2	12.8	13.8	13.1	13.1	11.4	16.1
Clear-Sky Solar Irradiance	taub	0.270	0.303	0.324	0.356	0.396	0.434	0.490	0.421	0.343	0.319	0.278	0.255	
	taud	1.995	2.008	2.051	2.134	2.241	2.222	2.064	2.247	2.444	2.327	2.065	1.958	
	Ebn at Noon	172	222	256	269	265	255	236	246	254	223	168	136	
	Edh at Noon	19	30	37	41	40	41	47	37	26	22	17	14	
All-Sky Solar Radiation	RadAvg	213	515	989	1438	1597	1814	1655	1337	829	412	224	136	
	RadStd	19	37	60	88	111	161	169	112	73	53	27	17	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page